

Homework #6. Due on Thursday, Oct 15th, 11:59pm on Canvas

Reading for this homework: Sections 3.1 and parts of 3.4.

Plan for next week (Oct 12, 14): 3.2 (the rank of a matrix), 4.2 (determinants of order n), possibly start 4.3 (properties of determinants).

Note: my plan is to define the determinants by the Leibniz formula

https://en.wikipedia.org/wiki/Leibniz_formula_for_determinants

instead of the cofactor expansion used in the book.

Problems:

For the first 2 problems do not assume anything about the rank except the following definitions:

- (1) Let $T : V \rightarrow W$ be a linear map where V and W are both finite-dimensional. The rank of T , denoted by $\text{rk}(T)$, is defined by

$$\text{rk}(T) = \dim(\text{Im}(T)) = \dim(T(V)).$$

- (2) If F is a field and $A \in \text{Mat}_{m \times n}(F)$, we define its rank $\text{rk}(A)$ by $\text{rk}(A) = \text{rk}(L_A)$ where $L_A : F^n \rightarrow F^m$ is the map of left multiplication by A , that is, $L_A(v) = Av$ for all $v \in F^n$.

Problem 1: Let $T : V \rightarrow W$ and $S : V \rightarrow W$ be linear maps.

- (a) Prove that $\text{Im}(T+S) \subseteq \text{Im}(T) + \text{Im}(S)$ and give an example showing that the equality may not hold.
- (b) Now assume that V and W are finite-dimensional. Prove that

$$\text{rk}(T+S) \leq \text{rk}(T) + \text{rk}(S).$$

Problem 2: Let $T : V \rightarrow W$ be a linear map and Z be a subspace of W .

- (a) Prove that $T^{-1}(Z)$ is a subspace of V . Here $T^{-1}(Z)$ is the preimage of Z , that is, $T^{-1}(Z) = \{v \in V : T(v) \in Z\}$.
- (b) Assume that V and W are finite-dimensional. Prove that

$$\dim(T^{-1}(Z)) \leq \dim(Z) + \dim(\text{Ker}(T))$$

Hint: Apply the rank-nullity theorem to a suitable map obtained from T by changing the domain.

Problem 3: In parts (a) and (b) let U, V, W be finite-dimensional vector spaces and $T : U \rightarrow V$ and $S : V \rightarrow W$ linear maps.

- (a) Prove that

$$\text{null}(ST) \leq \text{null}(S) + \text{null}(T)$$

where by definition $\text{null}(R) = \dim(\text{Ker}(R))$ for a linear map R .

Hint: You can express $\text{Ker}(ST)$ in a suitable way using preimages, so that the result would follow immediately from Problem 2.

- (b) Prove that $\text{rk}(ST) \geq \text{rk}(S) + \text{rk}(T) - \dim(V)$.
- (c) State and prove the analogue of (b) dealing with ranks of matrices.

Problem 4: Let F be a field and $m, n \in \mathbb{N}$. Recall from class that

- (i) $\mathcal{E}_{i,j}(\lambda)$ is the elementary row operation on the set $\text{Mat}_{m \times n}(F)$ which adds the j^{th} row multiplied by λ to the i^{th} row and does not change any other row (symbolically $\text{row}_i \mapsto \text{row}_i + \lambda \text{row}_j$).
- (ii) $\mathcal{E}'_{i,j}(\lambda)$ is the elementary column operation which adds the i^{th} column multiplied by λ to the j^{th} column and does not change any other column ($\text{col}_j \mapsto \text{col}_j + \lambda \text{col}_i$).
- (iii) $E_{ij}(\lambda)$ is the elementary (square) matrix (of suitable size) which has 1's on the diagonal, λ in the (i, j) -position and 0 everywhere else. Thus, $E_{ij}(\lambda) = I_k + \lambda e_{i,j}$ where k is the size of the matrix and I_k is the identity $k \times k$ matrix.

Prove (part of) Lemma 11.3 from class: for any $A \in \text{Mat}_{m \times n}(F)$ we have

- (a) $(\mathcal{E}_{i,j}(\lambda))(A) = E_{i,j}(\lambda)A$ and
- (b) $(\mathcal{E}'_{i,j}(\lambda))(A) = AE_{i,j}(\lambda)$,

that is, applying $E_{i,j}(\lambda)$ (resp. $\mathcal{E}'_{i,j}(\lambda)$) is the same as multiplying by $E_{i,j}(\lambda)$ on the left (resp. on the right).

Note: The notation $E_{i,j}(\lambda)$ is slightly ambiguous as it does not specify the size of the matrix. In particular, if $m \neq n$, then $E_{i,j}(\lambda)$ denotes matrices of different sizes in (a) and (b) above.

Problem 5: Recall two other types of elementary transformations introduced in class: $\mathcal{F}_{i,j}$ (swapping the i^{th} and j^{th} rows) and $\mathcal{D}_i(\lambda)$ (multiplying the i^{th} row by some nonzero $\lambda \in F$). Prove that $\mathcal{F}_{i,j}$ can be expressed as a composition of 3 operations of the form $\mathcal{E}_{k,l}(\lambda)$ and 1 operation of the form $\mathcal{D}_k(\lambda)$ (in a suitable order and for suitable k, l and λ).

Problem 6: In Lecture 11 we explained why every matrix A can be reduced to a matrix R in RREF (reduced row echelon form) using elementary row operations. The ultimate goal of this problem is to prove that such a matrix R is unique. Given $A \in \text{Mat}_{m \times n}(F)$, we define its row space $\text{Rowspace}(A)$ to be the subspace of F^n spanned by the rows of A .

- (1) Suppose that $A, B \in \text{Mat}_{m \times n}(F)$ are row equivalent (that is, can be obtained from each other by a sequence of elementary row operations). Prove that $\text{Rowspace}(A) = \text{Rowspace}(B)$. **Note:** The converse of (1) is also true, but requires more work to prove.

Now let $R \in \text{Mat}_{m \times n}(F)$ be a matrix in RREF. Our goal is to show that R can be recovered from its rowspace $\text{Rowspace}(R)$. Together with (1) this implies that any A is row equivalent to a UNIQUE matrix in RREF.

- (2) Let $e_1, \dots, e_n$ be the standard basis of F^n (which is viewed here as the space which naturally contains $\text{Rowspace}(R)$). For each $0 \leq k \leq n$ let $V_k = \text{Span}(e_1, \dots, e_k)$ be the span of the first k elements of the standard basis (in particular, $V_0 = \{0\}$ and $V_n = F^n$).

Now take $1 \leq k \leq n$. Prove that $V_k \neq V_{k-1} \iff R$ has a pivot in the k^{th} column. Thus, the set of columns of R which contain pivots is determined by $\text{Rowspace}(R)$.

- (3) Let $i_1 < i_2 < \dots < i_p$ be the indices of columns of R which contain pivots. Note that since R is in RREF, this means that the first p rows of R are nonzero, and the remaining rows are zero.

Let $T_0 : F^n \rightarrow F^p$ be the map which simply removes the j^{th} component for each $j \notin \{i_1, \dots, i_p\}$ (and preserves the order of the remaining components). For example, if $n = 5$, $p = 2$, $i_1 = 2$ and $i_2 = 4$, we have $T_0((a_1, a_2, a_3, a_4, a_5)) = (a_2, a_4)$.

Let $T : \text{Rowspace}(A) \rightarrow F^p$ be the restriction of T_0 to $\text{Rowspace}(R)$. Prove that T is an isomorphism and for all $1 \leq i \leq p$ we have $T(\text{row}_i(R)) = \varepsilon_i$, where $\{\varepsilon_1, \dots, \varepsilon_p\}$ is the standard basis of F^p . Thus, $\text{row}_i(R)$ is the unique element of $\text{Rowspace}(R)$ which is mapped to ε_i under T . Since the indices $i_1, \dots, i_p$ (and hence also the number p) are determined by $\text{Rowspace}(R)$ by (2) and all rows of R below the p^{th} row are zero, it follows that each row of R can be recovered from $\text{Rowspace}(R)$, as desired.

Problem 7: Let $A = \begin{pmatrix} 3 & -2 \\ 2 & -1 \end{pmatrix}$. Reduce A to the identity matrix only using operations of the form $\mathcal{E}_{12}(\lambda)$ and $\mathcal{E}_{21}(\lambda)$ for some integer λ . Then use Problem 4 to explicitly write A as a product of matrices of the form $E_{12}(\lambda)$ and $E_{21}(\lambda)$.

Problem 8: Let $n \in \mathbb{N}$ and F a field.

- (a) Let $A \in \text{Mat}_n(F)$ be an invertible matrix. Prove that the transposed matrix A^t is also invertible and $(A^t)^{-1} = (A^{-1})^t$.

- (b) Let $A \in \text{Mat}_n(F)$ be such that $A^k = 0$ for some $k \in \mathbb{N}$ (such matrices are called *nilpotent*). Prove that the matrix $B = I_n + A$ is invertible (as before, I_n is the identity matrix) and find an explicit formula for the inverse of A .

Do not use any facts about inverses except the definition. In (b) you may want to start with $k = 2$ (possibly followed by $k = 3$) before figuring out the general formula.